

Robust and Generalizable Proactive Face-Swapping Defense via Semantic Gradient Divergence

Appendix

Appendix Roadmap

This appendix provides supplementary evidence and implementation details for SGD-Guard. Section A presents an observational study showing that intentional attribute distortions disrupt face-swapping outputs more effectively than common visible edits such as watermarks, expression edits, and style changes. Section B analyzes the sensitivity of our method to the feature-gallery retrieval size k and justifies the default setting using qualitative and quantitative results in Fig B1 and Table B1. Section C provides additional qualitative results on robustness to post-processing, including JPEG compression, diffusion-based purification, and strong transformations, and on generalization under real-world conditions, resolution shifts, and cross-dataset settings in Figs. C1–C9. Section D details the complete offline and online procedures of SGD-Guard to facilitate reproducibility in Algorithms 1–2. Section E discusses limitations and outlines future directions.

A Results of Observational Experiments

A.1 Observational Study on Attribute Distortion in Deepfake Outputs

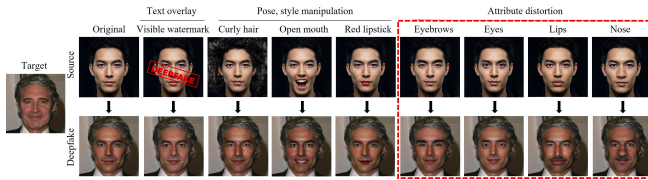


Figure A1: Observational analysis of deepfake output distortion. While existing manipulations such as text overlay and facial expression modifications have little impact on the resulting deepfakes, intentional attribute distortions lead to noticeable alterations in the generated outputs.

To study proactive defenses against face-swapping deepfakes, we applied several existing image manipulation techniques to the source images and analyzed the resulting swapped outputs. Specifically, we conducted experiments by adding visible watermarks to the source images, editing facial expressions using StyleCLIP [Patashnik *et al.*, 2021], and modifying overall style with CLIP2Protect [Shamshad *et al.*, 2023]. However, these manipulations had little to no impact

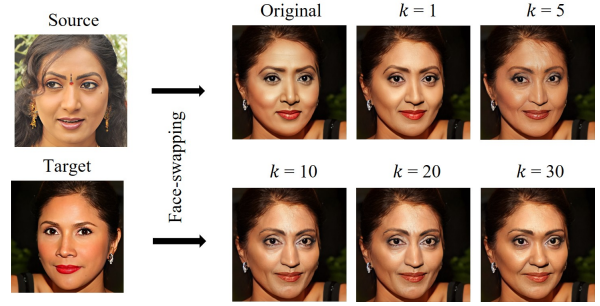


Figure B1: Effect of the feature count k on deepfake attribute distortion.

on the final deepfake results. To better understand how facial attribute manipulations affect face-swapping deepfakes, we further examined identity feature representations. In particular, we focused on a face recognition model that encodes faces into high-dimensional identity embeddings. By deliberately altering facial attributes that strongly influence these embeddings, we observed noticeable degradations in the resulting deepfake quality. As shown in Figure A1, such attribute-driven manipulations can disrupt identity consistency in the generated outputs more effectively than visible watermarks or expression/style edits.

B Evaluating Hyperparameter Influence

B.1 Sensitivity to the retrieval size k

In this section, we analyze the sensitivity of our method to the retrieval size k , which controls the top- k /bottom- k sample selection per attribute from the feature gallery. These retrieved subsets are used to compute CLIP and identity centroids that are concatenated to form the joint anchors (A_{top} , A_{bot}), and thus k directly affects the anchor construction used in optimization. We conduct both qualitative and quantitative evaluations on deepfake outputs synthesized by REFace [Hussain *et al.*, 2022] using images from VGGFace2-HQ [Chen *et al.*, 2023]. We report both qualitative and quantitative results to justify the default choice used throughout the paper.

Qualitative analysis

Figure B1 indicates that the retrieval size k exhibits a clear peak behavior in attribute distortion. The distortion strength-

ens as k increases from small values and becomes most pronounced at $k=10$, suggesting that retrieving a moderate number of features yields a coherent identity-aware reference that aligns well with the text-conditioned objective. In contrast, when k is increased beyond 10, the visual distortion weakens: the retrieved set increasingly contains relatively less relevant features, which dilutes the critical optimization signal and reduces the effectiveness of text-driven attribute disruption. This observation supports selecting $k=10$ as a practical operating point that maximizes qualitative distortion while maintaining stable behavior.

k	Impact of k		
	PSNR $\ddagger$ ↓	DSR $\uparrow$	FLOPs (M) ↓
1	20.87	89.1	81.92
5	20.56	90.8	81.94
10	20.03	92.3	81.96
20	20.05	92.2	82.01
30	20.11	91.6	82.06

Table B1: Sensitivity to the retrieval size k in the feature gallery.

Quantitative analysis

For quantitative evaluation, we select 1,000 source images and generate 10,000 face-swapped outputs via random source–target identity pairing. Table B1 further confirms the above trend quantitatively. Increasing k from 1 to 10 consistently improves protection while strengthening disruption (DSR: 89.1 $\rightarrow$ 92.3; PSNR: 20.87 $\rightarrow$ 20.03), and the computational cost remains essentially unchanged (FLOPs: 81.92M $\rightarrow$ 81.96M). Notably, once k exceeds 10, performance decreases: both the protection success rate and distortion metrics slightly deteriorate (e.g., $k=20$: DSR 92.2, PSNR 20.05; $k=30$: DSR 91.6, PSNR 20.11), accompanied by a small FLOPs increase (up to 82.06M). These results highlight that overly large retrieval can be counterproductive, and we therefore use $k=10$ by default in all experiments.

C Extra Qualitative Results

C.1 Results of Post-processing Robustness Experiments

We provide additional qualitative results under challenging post-processing operations. In this appendix, all results are generated using source images from CelebA-HQ [Karras *et al.*, 2017] and face-swapped outputs synthesized by REFace [Hussain *et al.*, 2022]. JPEG compression commonly appears in real-world face-sharing pipelines, whereas diffusion-based purification (adaptive attack) and extreme photometric and geometric transformations can be applied by adversaries to suppress protective perturbations. Accordingly, we qualitatively compare the robustness of SGD-Guard under JPEG compression with varying quality factors, diffusion-based purification, and extreme photometric and geometric transformations. These settings are particularly important because post-processing can attenuate or remove perturbations, causing many proactive defenses to fail. [Liu *et al.*, 2019] The following figures qualitatively illustrate the outcomes under each condition.

Qualitative robustness to JPEG compression

Figure C1 evaluates whether the proposed protection remains effective under JPEG post-processing with varying quality factors. We observe that Anti-Forgery, CMUA, DF-RAP, and SCOL generally exhibit weakened protection after compression, leading to face-swapping outputs that are only mildly disrupted. FaceShield shows a certain degree of resilience to JPEG compression; however, its overall protection effect becomes limited, and the resulting deepfakes remain relatively plausible. DiffAM is comparatively robust since it relies on makeup-like edits, but this comes with a noticeable discrepancy between the protected and original source images, indicating a larger appearance shift at the input. SCOL retains some protection and compression tolerance, yet the protected deepfakes do not appear substantially different from the originals in terms of perceived identity, limiting practical protection. In contrast, SGD-Guard consistently induces pronounced distortions in the resulting deepfake outputs across a wide range of JPEG qualities while keeping the protected source visually close to the original.

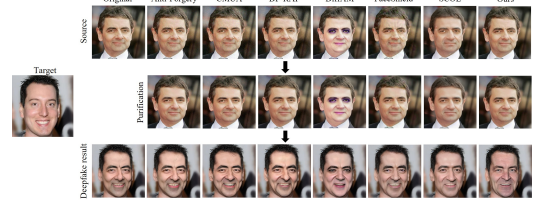


Figure C2: Visualization of protection performance against purification.

Qualitative robustness to purification

Figure C2 evaluates whether protective distortions remain effective after a diffusion-based purification [Nie *et al.*, 2022]. We observe that Anti-Forgery, CMUA, DF-RAP, FaceShield, and SCOL largely lose their protective distortions after purification, yielding face-swapping outputs that are only weakly disrupted. DiffAM is comparatively resilient to purification since it relies on makeup-like edits; however, the protected source image deviates noticeably from the original source, indicating a larger appearance shift at the input. In contrast, our method preserves strong protection under purification while maintaining low visual distortion on the protected source. This robustness stems from explicitly including the purification process in the optimization loop, encouraging perturbations that survive the purifier and continue to disrupt identity-critical cues after purification.

Qualitative robustness to extreme transformations

Figure C3 follows the settings in [Yuan *et al.*, 2022] and further analyzes robustness under extreme image transformations, including hue and saturation shifts, in-plane rotation, and pixel-value scaling. We observe that Anti-Forgery, CMUA, DF-RAP, FaceShield, and SCOL exhibit substantially weakened protective distortions under these aggressive transformations, yielding face-swapping outputs that are only weakly disrupted. DiffAM again remains relatively robust since it relies on makeup-like edits; however, this robustness

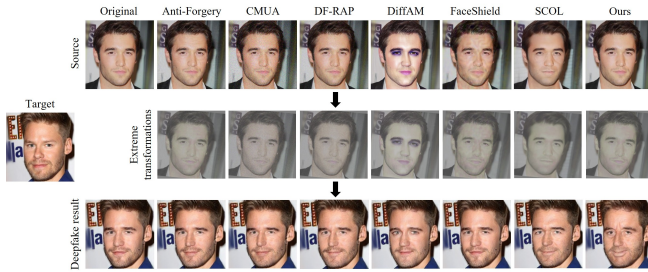


Figure C3: Visualization of protection performance under extreme transformations.

comes with a noticeable discrepancy between the protected and original source images. In contrast, our method remains robust under these conditions: by optimizing with semantic direction EOT, we focus updates on transformations that most strongly degrade transferability, producing perturbations that remain effective even under severe photometric and geometric shifts. Overall, Fig. C3 highlights that SD-EOT is critical for maintaining reliable protection beyond standard augmentation regimes.

C.2 Visual Results of Protected Images in Real-World Scenarios

We present additional qualitative results to assess whether SGD-Guard generalizes beyond standard, well-controlled inputs. We examine real-world challenging conditions such as low lighting, extreme poses, and occlusions, resolution changes from high-resolution inputs to severely downsampled inputs, and cross-dataset results on CelebA-HQ and VGGFace2-HQ. The following subsections compare deepfake outputs generated from original versus protected sources and highlight consistent distortions induced by SGD-Guard across these settings.

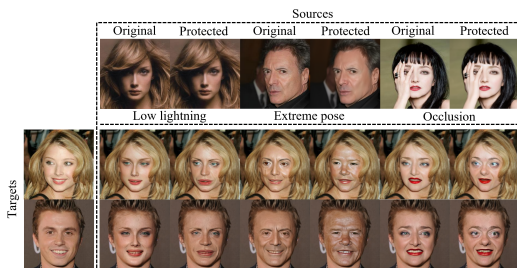


Figure C4: Deepfake outputs from original vs SGD-Guard protected images under real-world challenging scenarios.

Real-world challenging scenarios

Figure C4 evaluates SGD-Guard under visually challenging, real-world conditions that frequently degrade face analysis and face-swapping pipelines. We sample source images from CelebA-HQ that exhibit low lighting, extreme head poses, and occlusions, where identity cues become ambiguous, partially missing, or entangled with nuisance factors. Despite these appearance changes, SGD-Guard consistently yields deepfake outputs with pronounced distortions compared to

using the original sources, indicating that the protection effect generalizes beyond clean, canonical inputs.

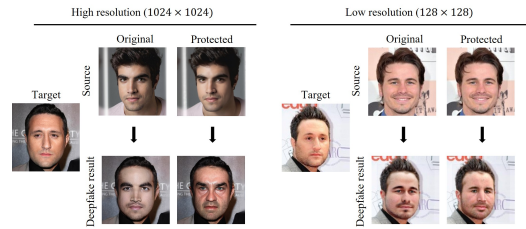


Figure C5: Qualitative robustness across image resolutions.

Qualitative generalization across image resolutions

Figure C5 evaluates the robustness of our protection under both high- and low-resolution settings by comparing deepfake outputs generated from original versus protected sources. For the *high-resolution* setting, we use CelebA-HQ images at 1024×1024 [Karras *et al.*, 2017]. For the *low-resolution* setting, we use CelebA images [Liu *et al.*, 2015] and construct 128×128 inputs via center cropping. Across both regimes, SGD-Guard consistently induces pronounced distortions in the resulting deepfake outputs, demonstrating that our protection does not rely on a specific input resolution and generalizes to both high-detail and severely downsampled scenarios.

Effectiveness of SGD-Guard on various datasets

We provide additional qualitative results in Figs. 6–9 to demonstrate the effectiveness of SGD-Guard across datasets and subject groups. Specifically, Figs. 6 and 7 report results on CelebA-HQ for female and male subjects, respectively, while Figs. 8 and 9 present results on VGGFace2-HQ for female and male subjects, respectively. Overall, SGD-Guard consistently induces pronounced distortions in the resulting deepfake outputs across both datasets and genders, suggesting robust generalization beyond a single dataset/domain.

D Additional Implementation Details

D.1 Full Procedure of SGD-Guard (Algorithm)

E Discussion

E.1 Limitations

Our method necessarily relies on a feature gallery to provide a stable reference space for identity-aware protection, which helps improve both consistency and robustness. However, this design also imposes a practical constraint because the system must always keep the feature gallery available at inference time, and it occupies approximately 72 MB in our implementation. In addition, protection strength can depend on how well the gallery covers the feature space. If the gallery contains too few features and lacks sufficient diversity, it may fail to represent a reliable reference distribution, which can reduce protection effectiveness. Developing more compact or gallery-free alternatives without sacrificing robustness remains an important direction for future work.

Algorithm 1 Offline generalized feature gallery build

Input: dataset $D = \{(x_o, y_o)\}_{o=1}^M$, where x_o is a gallery face image and y_o is its identity index; CLIP encoder E_{clip} ; frozen surrogates ArcFace($\cdot$), FaceNet($\cdot$); trainable parameters ϕ ; epochs E ; minibatch $\mathcal{B}$

Output: feature gallery $\mathcal{G}$

```
1: /* Train semantic iterative identity refinement */
2: Initialize  $\phi$ 
3: for epoch = 1 to  $E$  do
4:   Sample a minibatch  $\mathcal{B} \subset D$ 
5:    $e_1 \leftarrow [\text{ArcFace}(x_o)]_{(x_o, y_o) \in \mathcal{B}}$ 
6:    $e_2 \leftarrow [\text{FaceNet}(x_o)]_{(x_o, y_o) \in \mathcal{B}}$ 
7:    $y \leftarrow [y]_{(x_o, y_o) \in \mathcal{B}}$ 
8:    $z_1^0 \leftarrow e_1, z_2^0 \leftarrow e_2, R_1 \leftarrow 0, R_2 \leftarrow 0$ 
9:    $m^0 \leftarrow 0; n \leftarrow 0$ 
10:  repeat
11:     $n \leftarrow n + 1$ 
12:     $z_1^n \leftarrow \text{RBT}_1(z_1^{n-1}; \phi), z_2^n \leftarrow \text{RBT}_2(z_2^{n-1}; \phi)$ 
13:     $d_c^n \leftarrow \|z_1^n[c]\| - \|z_2^n[c]\|, m_c^n \leftarrow 1[d_c^n > H(d^n)]$ 
14:     $m^n \leftarrow [m_c^n]_{c=1}^{512}, \tilde{m}^n \leftarrow 1 - m^n$ 
15:     $R_1 \leftarrow R_1 + (\tilde{m}^n \odot z_1^n), R_2 \leftarrow R_2 + (\tilde{m}^n \odot z_2^n)$ 
16:     $z_1^n \leftarrow m^n \odot z_1^n, z_2^n \leftarrow m^n \odot z_2^n$ 
17:  until  $m^n = m^{n-1}$ 
18:   $\{I_u\}_{u \in \mathcal{B}} \leftarrow \text{Linear}(\text{concat}(R_1, R_2)); \phi$ 
19:  /* Compute gallery objective */
20:  Compute minibatch means/covariances  $(\bar{z}_1, \Sigma_1)$  from  $\{z_1^n\}_{u \in \mathcal{B}}$  and  $(\bar{z}_2, \Sigma_2)$  from  $\{z_2^n\}_{u \in \mathcal{B}}$ 
21:   $\mathcal{L}_{\text{statistic}} \leftarrow \mathcal{L}_{\text{statistic}}(\bar{z}_1, \Sigma_1, \bar{z}_2, \Sigma_2)$ 
22:   $\mathcal{L}_{\text{contrast}} \leftarrow \mathcal{L}_{\text{contrast}}(\{z_1^n\}_{u \in \mathcal{B}}, \{z_2^n\}_{u \in \mathcal{B}})$ 
23:   $\mathcal{L}_{\text{classification}} \leftarrow \mathcal{L}_{\text{classification}}(\{I_u\}_{u \in \mathcal{B}}, \{y_u\}_{u \in \mathcal{B}})$ 
24:   $\mathcal{L}_{\text{gallery}} \leftarrow \mathcal{L}_{\text{statistic}} + \mathcal{L}_{\text{contrast}} + \mathcal{L}_{\text{classification}}$ 
25:  Update  $\phi \leftarrow \phi - \alpha \nabla_{\phi} \mathcal{L}_{\text{gallery}}$ 
26: end for
27: /* Build feature gallery */
28: for  $o = 1$  to  $M$  do
29:    $C_o \leftarrow E_{\text{clip}}(x_o)$ 
30:    $I_o \leftarrow \text{identity refinement}(x_o; \phi)$  /* lines 5–18 forward */
31:   Add  $(C_o, I_o)$  to  $\mathcal{G}$ 
32: end for
33: return  $\mathcal{G}$ 
```

In addition, our current protection is designed and validated primarily at the single-image level. This means we do not explicitly enforce temporal consistency across consecutive frames. In video settings, attackers may exploit frame-to-frame redundancy, and small per-frame variations can accumulate into visible flicker or inconsistent protection over time.

E.2 Future Work

We aim to extend our current image-level protection to video-level deepfake protection, where maintaining temporal coherence is essential for stable and reliable defense across long sequences. Despite the substantial speed-up over prior methods, our current pipeline still falls short of millisecond-level latency, which limits practical deployment in real-time scenarios. To address this, we plan to develop universal adversarial perturbations (UAP) that can be applied with minimal per-frame overhead while preserving strong protection. In particular, we will design a UAP that generalizes across diverse deepfake generation models by more consistently af-

Algorithm 2 Online Procedure of SGD-Guard

Input: face image x ; fixed encoders $E_{\text{clip}}, E_{\text{id}}$; feature bank $\mathcal{G}$ (offline); transform set $\mathcal{T}$; JPEG qualities $\mathcal{Q}$; precomputed semantic directions $\mathcal{U}$; purifier $\text{Purify}(\cdot)$; steps T ; bound ϵ ;

Output: protected image $\hat{x}$

```
1: /* Initialize */
2:  $\delta \leftarrow \mathbf{0}_z$ 
3:  $\hat{x} \leftarrow x$ 
4: for  $t = 1$  to  $T$  do
5:   /* Semantic distortion attack */
6:   Compute joint features  $z \leftarrow \Phi(E_{\text{clip}}(\hat{x}), E_{\text{id}}(\hat{x}))$ 
7:   Compute loss  $\mathcal{L}_{\text{SD-attack}}(z; \mathcal{G})$ 
8:   /* Semantic direction EOT */
9:   Sample a subset  $\{T_p\}_{p=1}^P \subset \mathcal{T}$ 
10:  Compute attack direction  $d \leftarrow \text{Dir}(\hat{x}, x)$ 
11:  for  $k = 1$  to  $K$  do
12:     $u_k \in \mathcal{U}$ 
13:     $s_k \leftarrow \text{Score}(u_k, d)$ 
14:  end for
15:   $Z \leftarrow \sum_{i=1}^K \exp(\beta s_i)$ 
16:   $\mathcal{L}_{\text{eot}} \leftarrow 0$ 
17:  for  $k = 1$  to  $K$  do
18:     $w_k \leftarrow \exp(\beta s_k) / Z$ 
19:     $\tilde{x}_k \leftarrow T_p(\hat{x})$ 
20:     $z_k \leftarrow \Phi(E_{\text{clip}}(\tilde{x}_k), E_{\text{id}}(\tilde{x}_k))$ 
21:     $\mathcal{L}_{\text{SD-eot}} \leftarrow w_k \cdot \mathcal{L}_{\text{SD-attack}}(z_k; \mathcal{G})$ 
22:  end for
23:  /* Sematic robust attack */
24:  Sample qualities  $q_j \subset \mathcal{Q}$ 
25:   $\tilde{x} \leftarrow \text{JPEG}_{\text{real}}(\hat{x}; q_j)$ 
26:   $z \leftarrow \Phi(E_{\text{clip}}(\tilde{x}), E_{\text{id}}(\tilde{x}))$ 
27:   $\mathcal{L}_{\text{SR-JPEG}} \leftarrow \mathcal{L}_{\text{SD-attack}}(z; \mathcal{G})$ 
28:  /* Total loss and gradient computation */
29:   $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{SD-attack}} + \mathcal{L}_{\text{SD-EOT}} + \mathcal{L}_{\text{SR-JPEG}}$ 
30:   $g \leftarrow \nabla_{\delta} (\mathcal{L}_{\text{SD-attack}} + \mathcal{L}_{\text{SD-EOT}})$ 
31:   $+ \nabla_{\delta}^{\text{STE}} \mathcal{L}_{\text{SR-JPEG}}$ 
32:  /* PGD update */
33:   $\delta \leftarrow \delta - \alpha \cdot \text{sign}(g)$ 
34:   $\delta \leftarrow \text{clip}(\delta, -\epsilon, \epsilon)$ 
35:   $\hat{x} \leftarrow \text{clip}(x + \delta, 0, 1)$ 
36:  /* One step purification */
37:   $x_{\text{pur}} \leftarrow \text{Purify}(\hat{x})$ 
38:  end for
39:   $\hat{x} \leftarrow \text{clip}(x + \delta, 0, 1)$ 
40: return  $\hat{x}$ 
```

fecting shared feature representations extracted by different backbone networks. Moreover, we will explore how to identify and leverage semantic cues embedded in the UAP itself, using them as guidance to produce perturbations that induce more controllable and consistent feature distortion. We further target robustness against a broad range of post-processing and protection-bypassing attacks, ensuring that the protection cannot be easily removed or neutralized under realistic adversarial conditions.



Figure C1: Protection performance under JPEG compression with varying quality factors.

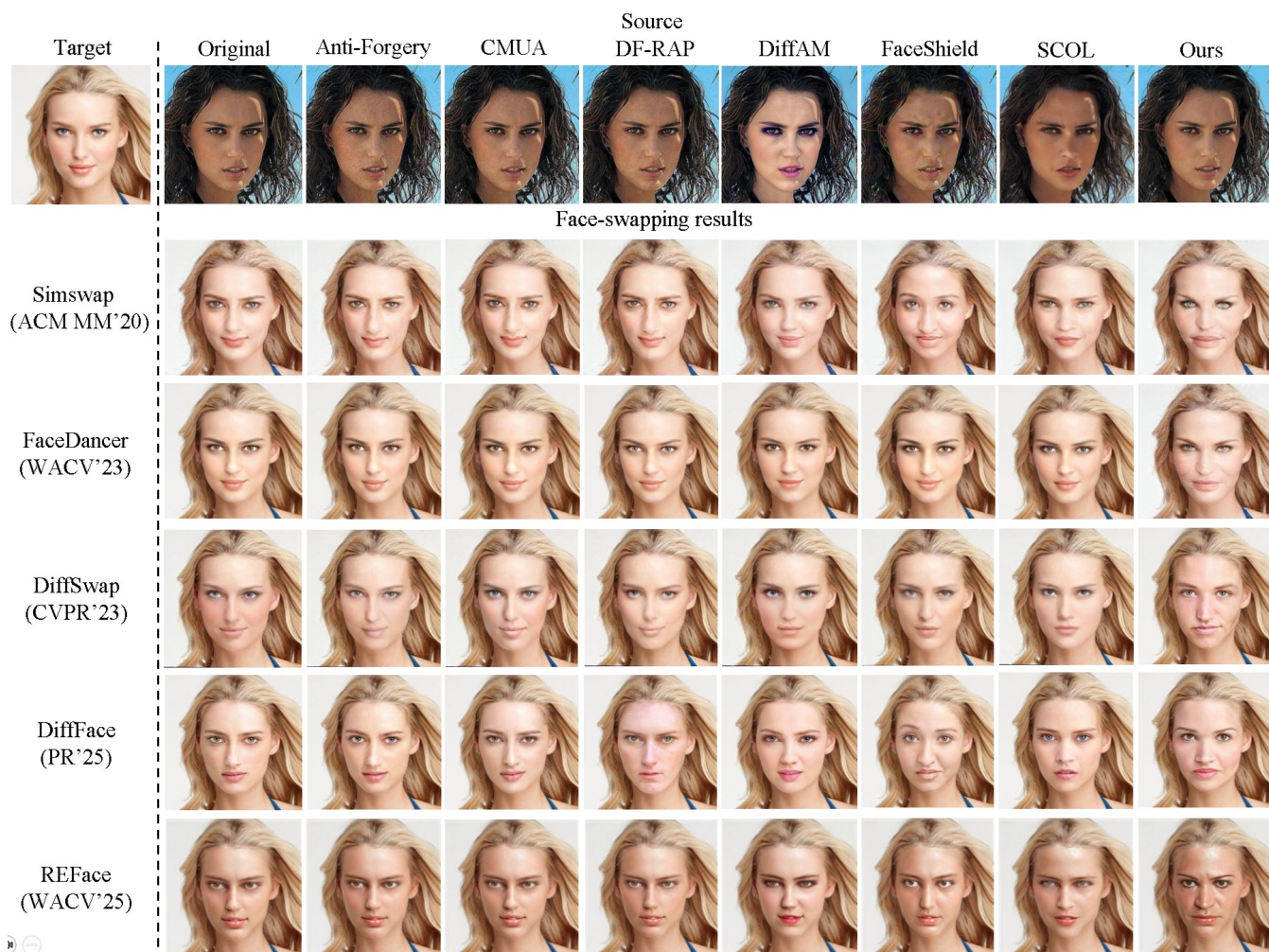


Figure C6: Visualization of existing methods and our proposed method on female images from the CelebA-HQ dataset.

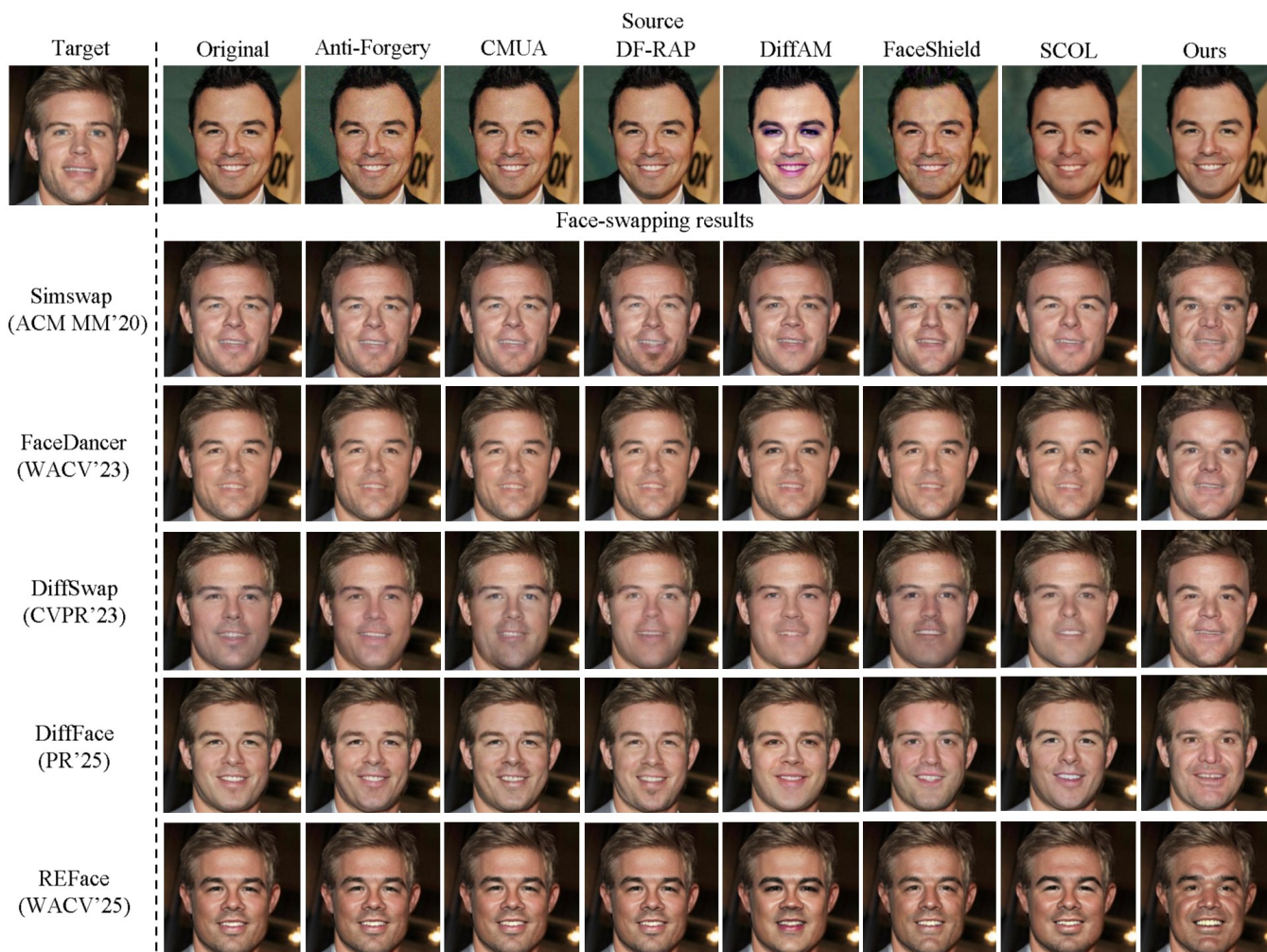


Figure C7: Visualization of existing methods and our proposed method on male images from the CelebA-HQ dataset.

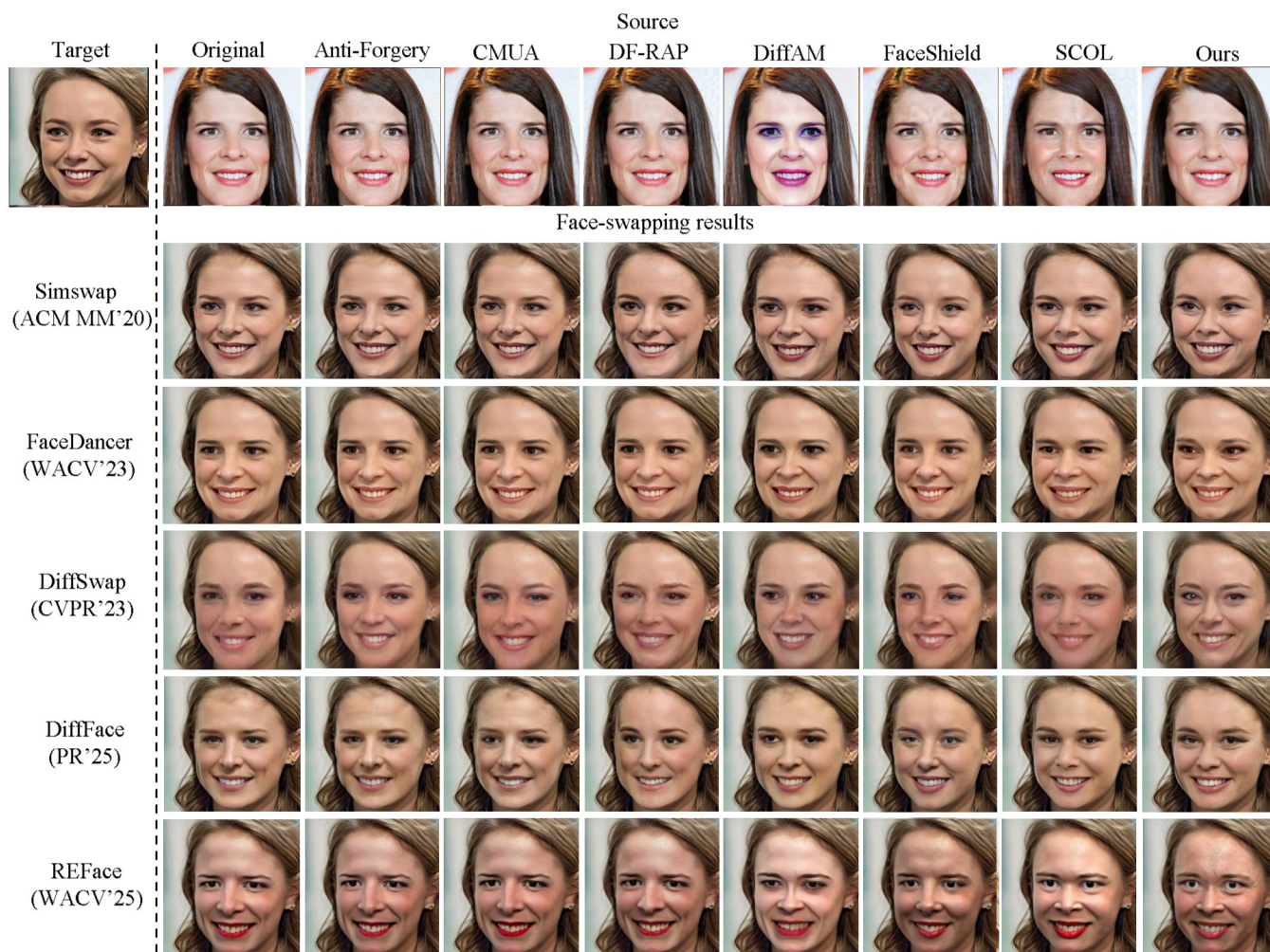


Figure C8: Visualization of existing methods and our proposed method on female images from the VGGFace2-HQ dataset.

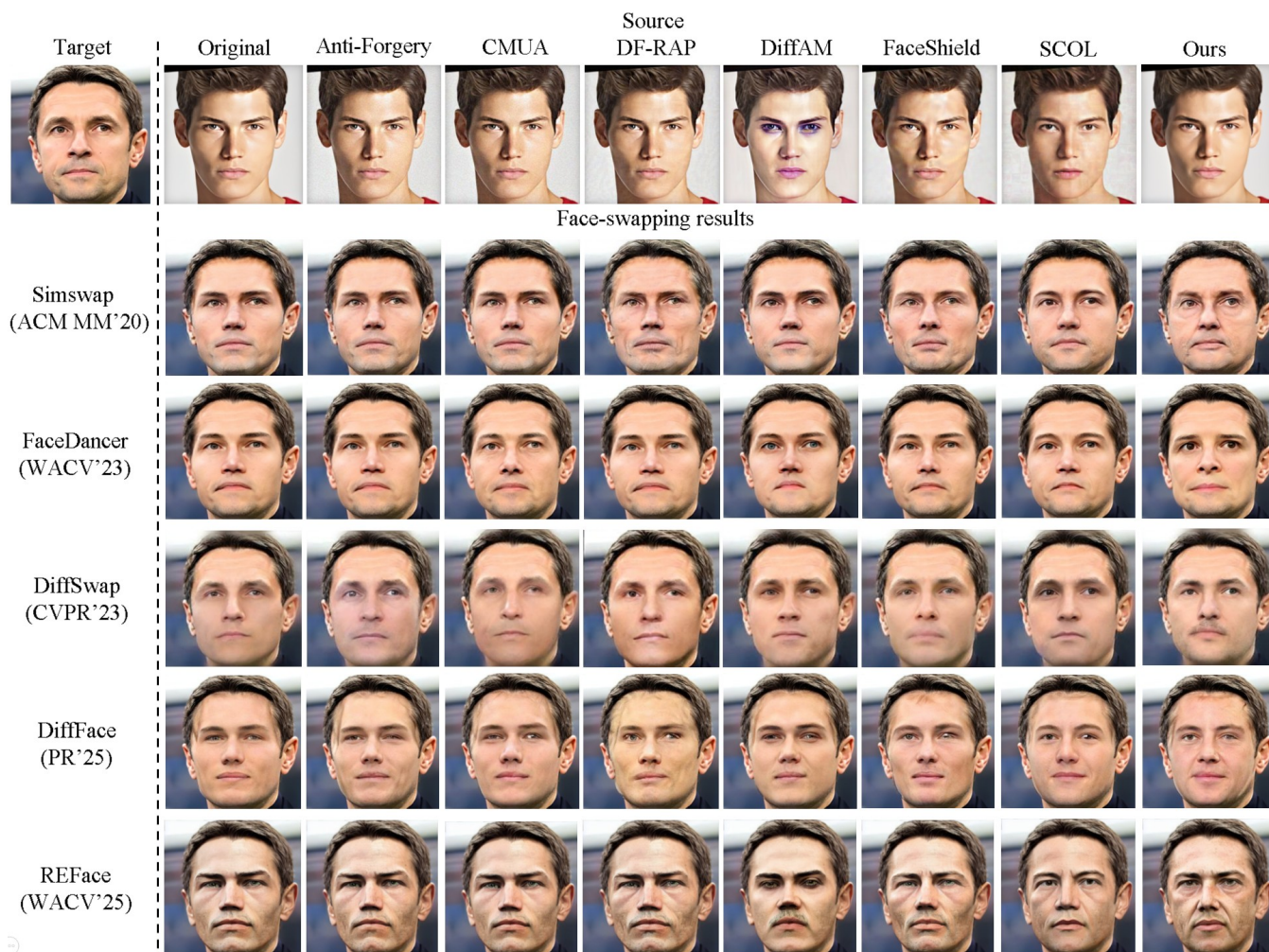


Figure C9: Visualization of existing methods and our proposed method on male images from the VGGFace2-HQ dataset.

References

- [Chen *et al.*, 2023] Xuanhong Chen, Bingbing Ni, Yutian Liu, Naiyuan Liu, Zhilin Zeng, and Hang Wang. Sim-swap++: Towards faster and high-quality identity swapping. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(1):576–592, 2023.
- [Hussain *et al.*, 2022] Shehzeen Hussain, Todd Huster, Chris Mesterharm, Paarth Neekhara, Kevin An, Malhar Jere, Harshvardhan Sikka, and Farinaz Koushanfar. Reface: Real-time adversarial attacks on face recognition systems. *arXiv preprint arXiv:2206.04783*, 2022.
- [Karras *et al.*, 2017] Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. *arXiv preprint arXiv:1710.10196*, 2017.
- [Liu *et al.*, 2015] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of the IEEE international conference on computer vision*, pages 3730–3738, 2015.
- [Liu *et al.*, 2019] Zihao Liu, Qi Liu, Tao Liu, Nuo Xu, Xue Lin, Yanzhi Wang, and Wujie Wen. Feature distillation: Dnn-oriented jpeg compression against adversarial examples. In *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 860–868. IEEE, 2019.
- [Nie *et al.*, 2022] Weili Nie, Brandon Guo, Yujia Huang, Chaowei Xiao, Arash Vahdat, and Anima Anandkumar. Diffusion models for adversarial purification. *arXiv preprint arXiv:2205.07460*, 2022.
- [Patashnik *et al.*, 2021] Or Patashnik, Zongze Wu, Eli Shechtman, Daniel Cohen-Or, and Dani Lischinski. Style-clip: Text-driven manipulation of stylegan imagery. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2085–2094, 2021.
- [Shamshad *et al.*, 2023] Fahad Shamshad, Muzammal Naseer, and Karthik Nandakumar. Clip2protect: Protecting facial privacy using text-guided makeup via adversarial latent search. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 20595–20605, 2023.
- [Yuan *et al.*, 2022] Zheng Yuan, Jie Zhang, and Shiguang Shan. Adaptive image transformations for transfer-based adversarial attack. In *European Conference on Computer Vision*, pages 1–17. Springer, 2022.